

QUANTTRADE AI

GLOBAL MONITOR

Product Requirements Document — v1.0

Bloomberg-Grade Geopolitical Intelligence & Live Market Risk Map

Attribute	Value
Document Owner	QuantTrade AI Product Team
Feature Name	Global Monitor — Full-Screen Macro Map
Version	1.0
Date	March 2026
Status	Ready for Engineering
Priority	P0 — Critical Path
Target Release	Q2 2026
Stakeholders	Product, Engineering, Design, Data

1. Executive Summary

QuantTrade AI Global Monitor is a production-grade, Bloomberg-inspired full-screen intelligence dashboard that transforms the existing mini-globe snapshot on the dashboard homepage into a dedicated, immersive geopolitical and macro-risk tracking experience. The feature directly upgrades the 404 placeholder page currently shown when users click "Open full global map" into a world-class live intelligence command center.

This PRD specifies the complete technical architecture, data integrations, UI/UX design system, and implementation roadmap required to ship at quanttrade.us production quality.

Core Value Proposition:

- Traders see geopolitical risk overlaid with market data on one map in real time
- Military, infrastructure, and conflict signals correlate directly with equities & commodities
- Bloomberg Terminal-like visual grammar — dark glass theme, information density, instant signal
- No \$30,000/year Bloomberg subscription required for macro context

2. Problem Statement & Strategic Context

2.1 Current State

The QuantTrade AI dashboard currently shows a decorative dark globe thumbnail on the right panel with the label "Sanctions, weather, and macro alerts." Clicking "Open full global map" routes to a 404-style page that reads: "The full-screen macro globe is almost ready." This creates a broken user experience and squanders a high-intent interaction from traders actively seeking macro intelligence.

2.2 Opportunity

Problem	Impact	Global Monitor Solution
Geopolitical signals not tied to trade thesis	Missed alpha on Iran/Russia/Taiwan risk events	Live conflict + military + sanctions overlay on map
News scattered across 100+ sources	Analysis latency of hours, not seconds	AI-synthesized briefs with geo-tagged events
No correlation between world events and assets	Manual cross-referencing wastes time	Click map marker → see impacted tickers instantly
Expensive OSINT platforms (\$3,000-\$30,000/yr)	Retail & boutique funds priced out	Free-tier and Pro tier access on QuantTrade
Dashboard globe is non-functional	Trust erosion, product incomplete feeling	Production-ready full-screen map experience

3. Goals, Non-Goals & Success Metrics

3.1 Goals

1. Ship production-ready Global Monitor page replacing current 404 placeholder
2. Implement live WebGL map with 20+ toggleable data layers
3. Integrate all free/freemium APIs from the WorldMonitor open-source project
4. Build AI-synthesized World Brief panel using Groq Llama 3.1
5. Enable ticker impact lookup: click any map event → see correlated equities/commodities
6. Deliver glass-themed, Bloomberg-grade UI that matches quanttrade.us design system
7. Achieve <2s initial map load, <500ms layer toggle, 60fps pan/zoom on modern hardware

3.2 Non-Goals (v1.0)

- Historical data playback (v2.0 roadmap)
- Push notification system (v2.0)
- Custom user-defined alert regions (v2.0)
- Mobile-first responsive map (v2.0 — desktop-first for v1.0)
- API access for programmatic consumption (v3.0)

3.3 Success Metrics

Metric	Target	Measurement Method
Map page adoption (DAU)	>40% of active users	Analytics: /global-monitor page views
Session duration on map	>3 minutes avg	Session tracking
Layer engagement rate	>60% toggle at least 1 layer	Click tracking per layer button
AI Brief generation latency	<3 seconds p95	Vercel Edge Function logs
Map initial load time	<2 seconds p90	Core Web Vitals / Lighthouse
User satisfaction (NPS)	+15 vs current dashboard	In-app survey, post-v1 launch
Ticker impact clicks	>25% of map-event interactions	Analytics: ticker panel opens

4. User Stories & Use Cases

4.1 Primary Personas

Persona	Description	Core Job-to-Be-Done
Discretionary Trader	Retail or boutique fund trader, 1-20 positions, monitors news	Understand geopolitical risk impact on my open positions before market open
Macro Analyst	Systematic desk or research arm, builds models	Correlate multi-signal intelligence (military + protests + infrastructure) with asset volatility
Risk Manager	Institutional desk, monitors portfolio tail risk	Get early warning on sanctions, conflict escalation before it hits price
Quant Researcher	Systematic strategy builder	Source geopolitical signals as features for factor models

4.2 User Stories

Story 1: Pre-Market Intelligence Briefing

As a discretionary trader, I want to open the Global Monitor at market open and immediately see a synthesized brief of overnight geopolitical developments, so that I can adjust my positions before the bell without reading 20 separate news sites.

- Acceptance: World Brief panel loads within 3 seconds with AI-generated summary
- Acceptance: Events are geo-tagged and visible on map as markers
- Acceptance: Each brief item links to impacted tickers (e.g., Iran event → OXY, CVX, USO)

Story 2: Event-to-Ticker Correlation

As a macro analyst, I want to click on any geopolitical event marker on the map and see a side panel listing correlated securities, commodities, ETFs, and options with recent price movement, so I can rapidly assess trade thesis validity.

- Acceptance: Click marker → side panel opens in <200ms

- Acceptance: Panel shows: event description, severity, affected tickers, 5-day price chart for each
- Acceptance: Ticker data pulls from existing QuantTrade data feeds

Story 3: Layer-Based Intelligence Filtering

As a risk manager, I want to toggle specific data layers (e.g., show only active conflicts + sanctions + energy infrastructure) to build a focused picture of geopolitical exposure for our energy sector book.

- Acceptance: All layers have a clearly labeled toggle button in a collapsible legend panel
- Acceptance: Layer state persists in URL params for sharing with team
- Acceptance: Layer toggle responds in <500ms with no map reload

Story 4: Country Instability Index

As a quant researcher, I want to see a real-time country instability score (0-100) for each monitored nation so I can use it as a signal input in my volatility models.

- Acceptance: Hover any country → tooltip shows CII score, component breakdown, trend
- Acceptance: Score updates every 5 minutes from live data
- Acceptance: CII methodology fully documented in API Docs page

5. Feature Specification

5.1 Dashboard Homepage Integration

The existing mini-globe snapshot (screenshot 1) requires the following production upgrades:

- Replace decorative dark globe with a live MapLibre GL mini-map rendering the top 3 active hotspots as animated markers
- Add a real-time event counter badge: e.g., "14 active alerts" pulsing in cyan
- "Open full global map" button links to /global-monitor route (no longer 404)
- Mini-map auto-rotates between 3 highest-severity active regions every 8 seconds
- Clicking the mini-map or the button opens the full-screen Global Monitor page

5.2 Full-Screen Global Monitor Page (/global-monitor)

5.2.1 Layout Architecture

The Global Monitor page follows a Bloomberg Terminal layout philosophy: maximum information density with zero wasted space. All panels are resizable and collapsible.

Zone	Location	Content	Width/Height
Top Bar	Full width, 48px	Time filters (1h/6h/24h/48h/7d), active layer count, data freshness indicator, region presets	100% / 48px
Left Panel	Left, collapsible	Layer controls, data source status, region jump buttons, legend	280px / full

Zone	Location	Content	Width/Height
Main Map	Center, fills remaining	WebGL interactive map with all data layers	Remaining / full
Right Panel	Right, collapsible	World Brief AI panel, Top Alerts feed, CII scores, Prediction Markets	380px / full
Ticker Impact Drawer	Bottom slide-up	Correlated securities when map marker clicked	100% / 320px
Status Bar	Bottom, 32px	API health indicators, last-updated timestamps, websocket status	100% / 32px

5.2.2 Map Engine

- Library: MapLibre GL JS (open-source Mapbox fork, no license fees)
- 3D Layer: deck.gl for WebGL-accelerated data visualization overlays
- Base Map Style: Custom dark glass QuantTrade theme built on CARTO Dark Matter tiles
- Projection: Globe projection for overview, Mercator for region zoom
- Performance: Tile caching, progressive loading, 60fps target on M2/i7+ hardware
- Clustering: Smart marker clustering at low zoom levels (deck.gl ClusterLayer)

5.2.3 Data Layers Specification

All 20+ layers are toggleable independently. Default-on layers are marked with [ON].

Layer Name	Default	Data Source	API/Endpoint	Update Freq
Active Conflict Zones	ON	ACLED / GDELT	api.acleddata.com + api.gdeltproject.org	15 min
Intelligence Hotspots	ON	Internal AI Engine	Internal signal aggregator	5 min
Breaking News Geo-Pins	ON	100+ RSS Feeds	Internal RSS proxy edge function	2 min
Social Unrest / Protests	ON	ACLED	api.acleddata.com/acledd/read	15 min
Sanctions Regimes	ON	OFAC / EU Data	Static + OFAC SDN list	Daily
Weather Severe Alerts	ON	NOAA / OpenMeteo	api.open-meteo.com	30 min
Military Bases (220+)	OFF	Static MIDB data	Bundled GeoJSON	Static
Live Military Flights	OFF	OpenSky Network	opensky-network.org/api	1 min
Naval Vessels (AIS)	OFF	VesselFinder / MarineTraffic	api.vesselfinder.com	5 min

Layer Name	Default	Data Source	API/Endpoint	Update Freq
Nuclear Facilities	OFF	NRC / IAEA Data	Bundled GeoJSON	Static
Undersea Cables	OFF	TeleGeography	Bundled GeoJSON	Static
Oil & Gas Pipelines	OFF	EIA / Public GIS	api.eia.gov	Static
AI Datacenters (111)	OFF	Static curated DB	Bundled GeoJSON	Weekly
Internet Outages	ON	Cloudflare Radar	api.cloudflare.com/radar	10 min
Satellite Fire Detection	OFF	NASA FIRMS/VIIRS	firms.modaps.eosdis.nasa.gov	3 hr
Earthquake Activity	ON	USGS	earthquake.usgs.gov/fdsnws	1 min
NASA Natural Events	ON	NASA EONET	eonet.gsfc.nasa.gov/api/v3	1 hr
Cyber Threat Actors (APT)	OFF	MITRE ATT&CK	attack.mitre.org (static)	Weekly
Prediction Markets	ON	Polymarket	gamma-api.polymarket.com	5 min
Economic Indicators	ON	FRED	api.stlouisfed.org/fred	Daily

6. API Integrations & Data Architecture

6.1 Required API Keys & Environment Variables

All keys are stored server-side in Vercel environment variables. Never exposed to the client.

Variable	Service	Purpose	Cost Tier
GROQ_API_KEY	Groq (Llama 3.1 8B)	World Brief AI synthesis, headline classification	Free tier: 14,400 req/day
UPSTASH_REDIS_REST_URL	Upstash Redis	Cross-user AI call dedup, Welford baseline storage, circuit breakers	Free tier: 10K req/day
UPSTASH_REDIS_REST_TOKEN	Upstash Redis	Auth token for Redis REST API	Included above
OPENSKY_USERNAME	OpenSky Network	Live military & civilian flight ADS-B tracking	Free with account

Variable	Service	Purpose	Cost Tier
OPENSKY_PASSWORD	OpenSky Network	Auth for higher rate limits (400 req/day vs 100)	Free with account
NASA_FIRMS_API_KEY	NASA FIRMS	VIIRS satellite thermal anomaly / fire detection	Free for non-commercial
ACLED_KEY	ACLED API	Armed conflict events, protests, social unrest data	Free for research/non-profit
EIA_API_KEY	US Energy Info Admin	Oil/gas pipeline data, energy market indicators	Free, no limit
FRED_API_KEY	Federal Reserve FRED	Economic indicators, interest rates, inflation data	Free, no limit
CLOUDFLARE_RADAR_TOKEN	Cloudflare Radar	Internet outage & BGP route leak detection	Free tier available
OPENROUTER_API_KEY	OpenRouter	AI fallback chain when Groq is unavailable	Pay-per-token

6.2 Free APIs (No Key Required)

- GDELT Project: api.gdeltproject.org/api/v2/doc — Global news event database, 15-min updates
- USGS Earthquake: earthquake.usgs.gov/fdsnws/event/1 — Real-time seismic events
- NASA EONET: eonet.gsfc.nasa.gov/api/v3/events — Natural disaster events
- Polymarket: gamma-api.polymarket.com — Geopolitical prediction markets
- OpenStreetMap Nominatim: nominatim.openstreetmap.org — Reverse geocoding for news geo-tagging
- OpenMeteo: api.open-meteo.com/v1 — Weather data, severe weather alerts
- CARTO Dark Matter Tiles: basemaps.cartocdn.com — Dark map tiles (free for non-commercial)
- IP-API: ip-api.com — IP geolocation for user context

6.3 Edge Function Architecture

Following the WorldMonitor architecture pattern, the Global Monitor uses Vercel Edge Functions as a lightweight API proxy layer. This keeps all API keys server-side, prevents CORS issues, enables cross-user Redis caching, and provides circuit breaker resilience.

Edge Function	Path	Responsibility
RSS Aggregator	/api/news/rss	Proxies 100+ RSS feeds, geo-tags with 74-hub database, classifies threats
GDELT Adapter	/api/news/gdelt	Normalizes GDELT quad-categories to QuantTrade event schema
Flight Tracker	/api/military/flights	OpenSky ADS-B proxy with military ICAO callsign filter
Vessel Monitor	/api/military/vessels	VesselFinder AIS proxy with naval vessel classification
Earthquake Feed	/api/hazards/earthquakes	USGS FDSN proxy with M4.0+ filter
NASA Events	/api/hazards/eonet	NASA EONET natural events normalized feed

Edge Function	Path	Responsibility
NASA FIRMS Fires	/api/hazards/fires	VIIRS thermal anomaly data, 24hr window
ACLED Unrest	/api/intel/acled	Armed conflict + protest events with caching
Internet Outages	/api/infra/outages	Cloudflare Radar BGP prefix withdrawal monitor
AI World Brief	/api/ai/brief	Groq Llama 3.1 synthesis with Redis dedup, 15-min cache
AI Classifier	/api/ai/classify	Hybrid keyword + LLM threat classifier, 24hr Redis cache
Instability Index	/api/intel/cii	Country Instability Index computation from all signal sources
Focal Points	/api/intel/focal	Multi-signal convergence detection, 5-min cache
Prediction Markets	/api/markets/polymarket	Geopolitical market odds, tag-filtered, 5-min cache
Economic Indicators	/api/markets/fred	FRED macro indicators, daily cache
Temporal Baseline	/api/intel/anomaly	Welford streaming baseline, z-score anomaly detection
Signal Aggregator	/api/intel/signals	Master signal fusion endpoint for map overlay data

6.4 Threat Classification Pipeline

Every news item passes through a two-stage pipeline identical to WorldMonitor's proven architecture:

- 8. Keyword Classifier (instant, client-side): Pattern-matches ~120 threat keywords organized by severity tier. Returns confidence score immediately. Zero latency.
- 9. LLM Classifier (async, edge function): Groq Llama 3.1 8B at temperature=0 fires in background. Results cached in Redis (24hr TTL) keyed by headline SHA-256 hash. Overrides keyword result only when LLM confidence is higher.

Severity Tiers: CRITICAL (red) → HIGH (orange) → MEDIUM (yellow) → LOW (blue) → INFO (gray)

Threat Categories: conflict, terrorism, cyber, disaster, energy, economic, political, military

6.5 Country Instability Index (CII) Formula

Component	Weight	Data Source	Notes
Baseline Structural Risk	40%	Static config per country	Pre-configured fragility scores, conflict floors (Ukraine ≥55, Syria ≥50)
Unrest Events	20%	ACLED	Log-scale for democracies, linear for authoritarian states; fatality boost
Security Activity	20%	OpenSky + VesselFinder	Military flights (3pts), vessels (5pts), foreign military presence (2x weight)
Information Velocity	20%	RSS + GDELT	News mention frequency × severity multiplier, log-scaled
Boosts	+variable	All sources	Hotspot proximity, focal point urgency, convergence zone multiplier

7. UI/UX Design System

7.1 Design Philosophy

The Global Monitor adopts a Bloomberg Terminal-inspired glass morphism design system: dark backgrounds with semi-transparent frosted glass panels, cyan/teal accent colors that signal live data, minimal chrome, and maximum information density. Every pixel serves a purpose.

7.2 Color Palette

Token	Value	Usage
--bg-void	#020408	Map background, outermost shell
--bg-base	#0A0E1A	Primary panel backgrounds
--bg-elevated	#0F1629	Cards, input fields
--bg-glass	rgba(15,22,41,0.7)	Glass panel overlay on map
--glass-border	rgba(0,212,255,0.15)	Glass panel borders (frosted edge)
--accent-primary	#00D4FF	Primary interactive elements, live indicators
--accent-success	#00FFB3	Positive data, bullish signals, stable status
--accent-warning	#FFB800	Elevated alerts, degraded data sources
--accent-critical	#FF2D55	High alert markers, critical events
--accent-nuclear	#FFD700	Nuclear facility markers
--text-primary	#E8F4F8	Primary readable text
--text-secondary	#7A8FA6	Secondary labels, timestamps
--text-dim	#3A4A5C	Disabled states, very secondary

7.3 Typography

- Primary Font: JetBrains Mono — for data values, coordinates, tickers, numbers
- UI Font: Inter — for labels, descriptions, panel titles
- Display Font: Space Grotesk — for section headers, page title
- All fonts loaded via Google Fonts CDN with font-display: swap

7.4 Map Marker Design System

Marker Type	Shape	Color	Animation	Size
High Alert Event	Circle with outer ring pulse	#FF2D55	Expanding ring, 2s loop	24px + 48px ring
Elevated Event	Circle	#FFB800	Slow pulse, 3s loop	18px
Monitoring Event	Circle	#00D4FF	Static	14px

Marker Type	Shape	Color	Animation	Size
Military Base	Triangle	#7A8FA6	Static	16px
Nuclear Facility	Star/Diamond	#FFD700	Subtle glow	20px
Datacenter	Square	#9B59B6	Static	16px
Conflict Zone	Heat polygon	#FF2D55 fill 0.3 opacity	None	Area fill
Breaking News	Location pin	#00D4FF	Drop animation on appear	22px
Protest/Unrest	Crowd icon	#FFB800	Static	18px

7.5 Glass Panel Component Spec

All side panels use the following CSS architecture (to be implemented in the QuantTrade design system):

- Background: rgba(10, 14, 26, 0.85)
- Backdrop Filter: blur(20px) saturate(180%)
- Border: 1px solid rgba(0, 212, 255, 0.15)
- Border Radius: 12px
- Box Shadow: 0 8px 32px rgba(0, 0, 0, 0.6), inset 0 1px 0 rgba(255,255,255,0.05)
- Header accent line: 2px solid gradient(#00D4FF → #00FFB3)

7.6 Interaction Patterns

- Layer toggle: Instant visual response, then data load with skeleton shimmer
- Marker click: Panel slides in from right (320ms ease-out cubic bezier)
- Map hover: Tooltip appears after 150ms debounce delay
- Time filter: Map animates to filter events, exiting markers fade out
- Region preset: Map flies to region with 1.5s animated transition
- AI Brief refresh: Spinner on button, streaming text response renders word-by-word

8. Technical Architecture

8.1 Tech Stack

Category	Technology	Rationale
Framework	Next.js 14+ (App Router)	QuantTrade existing stack, RSC for server data, edge functions
Map Engine	MapLibre GL JS 3.x	Open-source, no Mapbox license fees, full WebGL support
WebGL Overlays	deck.gl 8.x	GPU-accelerated 100K+ marker performance, ScatterplotLayer, HeatmapLayer

Category	Technology	Rationale
State Management	Zustand	Lightweight, map layer state, side panel state
Data Fetching	SWR + React Query	Stale-while-revalidate for live data, automatic background refresh
Styling	Tailwind CSS + CSS Variables	QuantTrade existing system + custom glass tokens
AI Runtime	Groq SDK + Vercel AI SDK	Streaming responses for World Brief panel
Caching	Upstash Redis (REST)	Cross-user AI dedup, Welford baselines, circuit breaker state
Charts	Recharts / Lightweight Charts	Ticker impact drawer price charts
WebSocket	Native WS / Pusher	Real-time marker appear animations, live feed updates
Build	Vite (inside Next.js) / Turbopack	Fast HMR in development
Deploy	Vercel Edge Network	Edge functions co-located with data sources

8.2 Data Flow Architecture

The Global Monitor follows a hub-and-spoke data architecture. All external APIs are contacted exclusively through Vercel Edge Functions. The client never calls external APIs directly.

- 10. Client boots → loads base map tiles from CARTO CDN
- 11. Client calls /api/intel/signals — master aggregated signal endpoint (5-min SWR cache)
- 12. Signal aggregator returns: hotspots, conflict zones, military data, events, CII scores
- 13. Client renders initial map state with deck.gl layers
- 14. Client calls /api/ai/brief for World Brief panel (15-min SWR cache)
- 15. Background workers poll individual layer endpoints on per-layer refresh intervals
- 16. WebSocket connection receives real-time breaking news events for immediate map pin drop
- 17. On marker click: client calls /api/markets/ticker-correlation?event={id} for ticker impact data

8.3 Caching Strategy

Data Type	Cache TTL	Cache Layer	Invalidation
AI World Brief	15 minutes	Upstash Redis (cross-user)	Manual refresh button; TTL expiry
Headline classification	24 hours	Upstash Redis (by headline hash)	TTL expiry only
ACLED events	15 minutes	Edge Function memory + Redis	TTL expiry
GDELT events	5 minutes	Edge Function memory	TTL expiry
OpenSky flights	60 seconds	Edge Function memory	TTL expiry
USGS earthquakes	60 seconds	Edge Function memory	TTL expiry

Data Type	Cache TTL	Cache Layer	Invalidation
NASA FIRMS fires	3 hours	Edge Function memory + Redis	TTL expiry
Polymarket odds	5 minutes	Edge Function memory	TTL expiry
FRED indicators	24 hours	Redis	Daily cron invalidation
CII scores	5 minutes	Computed client-side	On signal data refresh
Temporal baselines	90-day rolling window	Upstash Redis (Welford state)	Appended on each observation
Static GeoJSON layers	Indefinite	Vercel CDN edge cache	Git deploy invalidation

8.4 Performance Requirements

- First Contentful Paint (map tiles + skeleton): <1.5s on 10Mbps connection
- Time to Interactive (first data layer rendered): <3s
- Layer toggle response time: <500ms
- Marker click to panel open: <200ms (local state, no network)
- Sustained rendering: 60fps at 10,000 active markers, 30fps at 50,000
- Memory ceiling: <500MB RAM on map page
- Bundle size: <800KB gzipped for map page chunk

9. Implementation Roadmap

9.1 Phase 1 — Foundation (Week 1–2)

- Set up /global-monitor Next.js route, remove 404 placeholder
- Initialize MapLibre GL JS with custom QuantTrade dark glass tile style
- Implement basic deck.gl ScatterplotLayer for static marker rendering
- Create QuantTrade globe design system components (GlassPanel, MapMarker, LayerToggle)
- Set up Vercel Edge Function boilerplate with circuit breaker pattern
- Implement USGS earthquake feed edge function (simplest free API to validate stack)
- Connect mini-globe on dashboard to route, remove 404 path
- Deploy to staging environment

9.2 Phase 2 — Core Data Layers (Week 3–4)

- RSS feed aggregator edge function (100+ feeds, geo-tagging with 74-hub DB)
- GDELT news event adapter
- Keyword threat classifier implementation
- ACLED social unrest & armed conflict feed
- Conflict zone polygon rendering on map
- NASA EONET natural disaster events
- Breaking news WebSocket stream for real-time pin drops

- Time filter controls (1h / 6h / 24h / 48h / 7d)

9.3 Phase 3 — Intelligence Engine (Week 5–6)

- Groq AI World Brief edge function with Redis caching
- LLM threat classifier (async override of keyword classifier)
- Country Instability Index computation
- Focal point detection algorithm
- Temporal baseline anomaly detection (Welford algorithm in Redis)
- Signal aggregator master endpoint
- Geographic convergence detection ($1^\circ \times 1^\circ$ cell binning)
- Right panel: World Brief + Top Alerts + CII scores

9.4 Phase 4 — Advanced Layers & QuantTrade Integration (Week 7–8)

- OpenSky military flight tracking edge function
- NASA FIRMS satellite fire detection
- Cloudflare Radar internet outage monitoring
- FRED economic indicators integration
- Polymarket geopolitical prediction markets panel
- TICKER IMPACT FEATURE: Click map event → correlated equities panel
- Integrate QuantTrade existing stock data API into ticker impact drawer
- Undersea cables, pipelines, nuclear facilities static GeoJSON layers

9.5 Phase 5 — Polish & Production (Week 9–10)

- VesselFinder AIS naval tracking (requires paid API key)
- URL state persistence for shareable map views
- Region preset animations (fly-to transitions)
- Data freshness indicator panel (14-source health monitor)
- Layer state memory per user session
- Performance optimization: bundle splitting, tile preloading
- Accessibility audit and ARIA labels for all interactive elements
- Lighthouse score targets: Performance >85, Accessibility >90
- Production deploy + monitoring setup (Vercel Analytics + Sentry)
- Announce to users, replace 404 message in-app

10. Design Specification for Engineers

10.1 Global Monitor Page Layout (Wireframe Reference)

The following describes the pixel-level layout for Figma implementation and developer handoff:

Element	Position	Dimensions	CSS Class
Page wrapper	Full viewport	100vw × 100vh, overflow:hidden	.gm-root

Element	Position	Dimensions	CSS Class
Map canvas	z-index: 0, full page	100vw × 100vh	.gm-map-canvas
Top bar glass panel	top:0, left:0, full width	100vw × 48px	.gm-topbar
Left layer panel	top:48px, left:0	280px × calc(100vh-80px)	.gm-layers-panel
Right intel panel	top:48px, right:0	380px × calc(100vh-80px)	.gm-intel-panel
Ticker impact drawer	bottom:32px, left:0	100vw × 0 (320px when open)	.gm-ticker-drawer
Status bar	bottom:0, left:0	100vw × 32px	.gm-statusbar
Marker tooltip	Dynamic, follows cursor	max-width: 280px	.gm-tooltip
Event detail panel	Slides from right	400px × calc(100vh-48px)	.gm-event-panel

10.2 Right Intelligence Panel Sections

- 18. WORLD BRIEF: AI-synthesized 3-paragraph summary, refresh button, timestamp. Font: Inter 13px, streaming render.
- 19. TOP ALERTS: Scrollable feed of top 10 severity-ranked events. Each item: severity badge, headline, source tier, 1-line summary, geo-link to map.
- 20. COUNTRY INSTABILITY INDEX: Top 10 most-watched nations, each with: flag emoji, score bar 0-100, trend arrow (up/down/flat), click to zoom map to country.
- 21. PREDICTION MARKETS: Top 5 active geopolitical Polymarket contracts, probability %, 24hr change, direct link.
- 22. DATA FRESHNESS: Grid of 14 data source status dots (green/yellow/red), last-updated time, click for detail.

10.3 Ticker Impact Drawer Content

When user clicks any map event marker, the bottom drawer slides up with:

- Event summary card: title, severity, timestamp, source
- Affected Asset Categories: Energy, Defense, Agriculture, Currencies (auto-detected from event category)
- Correlated Tickers Grid: 4-6 tickers with logo, current price, 5d change, mini sparkline
- Relevant ETFs: e.g., for Middle East event → show OIH, XLE, USO, GLD
- "Add to Watchlist" button for each ticker (integrates with QuantTrade Watchlist feature)
- "Open in Markets" deep link to QuantTrade Markets page pre-filtered to sector

11. Risk Register & Mitigations

Risk	Probability	Impact	Mitigation
ACLED API free tier rate limit hit	Medium	High	Redis 15-min cache, circuit breaker, GDELT as fallback for conflict data
OpenSky free tier insufficient for real-time	High	Medium	Cache 60s, display data-age indicator, document limitation to users
Groq API latency >5s	Low	Medium	Async loading, show skeleton, fallback to OpenRouter, then browser T5 model
MapLibre GL performance on older hardware	Medium	Medium	Progressive layer loading, auto-disable layers at <30fps threshold, WebGL detection
CARTO tile CDN outage	Low	Critical	Fallback tile URL to OpenStreetMap Tile CDN
Data accuracy / false geopolitical alarm	Medium	High	Source tier labeling, confidence scores, disclaimer in UI, user reporting
Upstash Redis free tier quota (10K req/day)	Medium	Medium	Selective caching (AI calls only), upgrade to paid if needed at scale
VesselFinder API cost	N/A	Medium	Defer to v1.1, note as beta feature, use free AIS alternatives if found
Build complexity delays timeline	Medium	High	Ship phases independently, Phase 1-2 as MVP, phases 3-5 as enhancements

12. Open Questions & Decisions Required

#	Question	Owner	Target Date
1	What is the budget for paid API keys? (VesselFinder ~\$50/mo, paid Upstash ~\$20/mo)	Product Lead	Week 1
2	Should ACLED data be used for QuantTrade (they require attribution for non-profits)?	Legal	Week 1
3	Ticker impact correlation: use QuantTrade's existing data API or build new mapping?	Engineering Lead	Week 2
4	Should Global Monitor be gated behind Pro plan, or available on free tier?	Product Lead	Week 1
5	Which 20 countries should be in the default CII monitoring set for v1.0?	Product + Research	Week 2
6	Do we build our own 74-hub geo-tagging database or fork WorldMonitor's directly?	Engineering Lead	Week 2

#	Question	Owner	Target Date
7	WebSocket for live news: build on Vercel (limited duration) or use Pusher/Ably?	Engineering Lead	Week 3
8	Should AI World Brief be streaming (token-by-token) or batch (full paragraph)?	Product + UX	Week 2

13. Appendix: WorldMonitor Reference Implementation

13.1 Key Architectural Decisions to Adopt from WorldMonitor

- Browser-first compute: CII scoring, clustering, and anomaly detection run client-side. No backend compute dependency for core intelligence. Enables zero-server-cost scaling.
- Hybrid classifier pattern: Keyword instant → LLM async override. Never block UI on AI.
- Per-feed circuit breakers: 5-minute cooldowns prevent cascading failures. Always show stale data rather than blank panels.
- Intelligence gap reporting: Explicitly show what data is missing/stale. Do not hide source outages.
- Multi-signal correlation requirement: No single source triggers CRITICAL. Requires convergence.
- Welford streaming algorithm: Numerically stable mean/variance for anomaly detection without storing full history.

13.2 WorldMonitor Tech Stack Reference

- Map: MapLibre GL JS + deck.gl (WebGL) → QuantTrade adopts both
- AI: Groq Llama 3.1 8B + TensorFlow.js browser fallback → QuantTrade adopts same
- Cache: Upstash Redis REST API → QuantTrade adopts same
- Deploy: Vercel Edge Functions → QuantTrade already on Vercel
- RSS: Domain-allowlisted proxy edge function → QuantTrade builds identical pattern
- Data: TypeScript, Vite, 100+ feeds → QuantTrade uses Next.js (compatible)

13.3 QuantTrade-Specific Enhancements vs WorldMonitor

Feature	WorldMonitor	QuantTrade Global Monitor
Asset correlation	No ticker integration	Click event → correlated equities/ETFs/options panel
Portfolio overlay	Not present	Overlay user watchlist positions on map (Phase 2)
Market data	Prediction markets only	Full QuantTrade stock/options data integrated
Design system	Generic dark theme	Glass morphism Bloomberg-grade QuantTrade brand system
User context	Anonymous/public	Authenticated, personalized watchlist-aware map

Feature	WorldMonitor	QuantTrade Global Monitor
News feeds	Geopolitics/defense focus	Adds QuantTrade financial news feeds (CNBC, Bloomberg, Reuters markets)
Trade thesis	Not present	AI panel suggests trades based on event type and sector impact

END OF DOCUMENT

QuantTrade AI — Global Monitor PRD v1.0 — March 2026 — quanttrade.us
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